CAIRN Field Guide

How to send “I’m OK” and “I need help” when phones do not work.

Quick start — read this first

CAIRN sends short messages from radio to radio. It needs **no phone signal, no internet and no licence**. Other CAIRN or MeshCore radios (or a repeater) must be within range to hear you.

1. **Antennas on first.** Screw the antenna onto **each** radio before you give it power. A radio that sends with no antenna can be damaged.

2. **Plug in.** Plug the tablet, the CAIRN radio (the one with the little screen) and the small repeater radio into the USB hub. Plug a charged power bank into the hub. Wait one minute.

3. **Open the CAIRN page** on the tablet. Tap the screen to get past the opening picture. Press **Connect radio** and choose the radio.

4. **Say you are OK.** On the **Home** screen press **I’M OK**. Everyone in range sees your tag and where you are.

5 **In trouble? Call 911 first if you can.** If you cannot, press **I NEED HELP**, then **YES, SEND HELP**. It keeps calling every few minutes until you cancel.

Remember

Everything you send, including your position, can be read by anyone with a radio in range. Your tablet and radio need power: check the **Power** screen now and then.

No tablet? The CAIRN radio has its own small screen. Short press the button to change screens. On the **HELP** screen, hold the button, then hold it again within 5 seconds to send a help call.

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1. What CAIRN is, and what it is not

CAIRN is a pair of small radios and a tablet. It lets people near each other send short text messages, check in, and call for help when phones and the internet are down. It also works on a normal day for hikes, events and camping.

It can

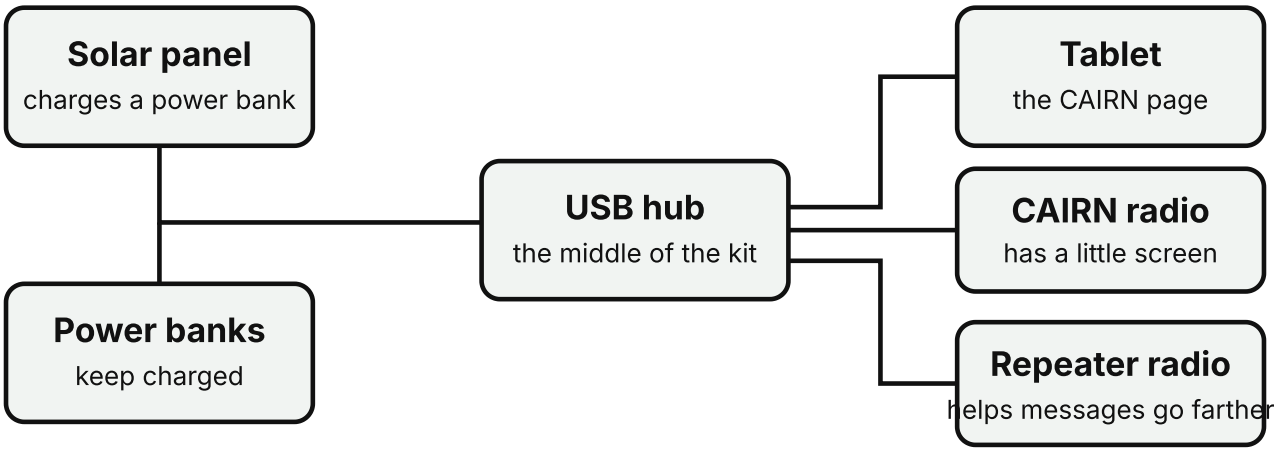
- Send “I’m OK” with your position
- Call for help, again and again, until cancelled
- Send short messages (up to 140 letters)
- Show who has reported, who needs help, and what is marked on the map
- Work with no internet at all (load maps ahead of time)

It cannot

- **Call 911** or reach emergency services by itself
- Reach people who have no radio in range
- Promise a message arrived
- Keep a secret on the Public channel

How far it reaches changes a lot. In a city, a few blocks. In the open, or with a repeater on a roof or hill, much farther. Height helps more than anything else: go up, go outside.

2. What is in the kit



Your kit card shows the exact cables. Keep the two radios' antennas **apart from each other and away from the hub and cables**: that gives the best range.

3. First-time setup (do this at home, with internet)

1. **Charge everything** — tablet, power banks, and the radio's own battery if it has one.

2. **Open the CAIRN page** in Chrome on the tablet. In the browser menu choose **Add to Home screen**. After that it opens with no internet.

3. **Load a map for your area.** Go to **Map**, pick your region in the list, download it, then press **Load a map file**. Maps are big, so use Wi-Fi.

4. **Try Practice.** At the top, press **Practice**. The page fills with pretend people so you can press every button safely. Press **Stop practice** when done.

5 **Check the antenna number.** Go to **Power**. The radio assumes a normal antenna (2 dBi). If you fit a different antenna, type the number printed on it in **Antenna gain** and press **Set**. This keeps the radio inside the legal power limit.

6 **Choose where "I'm OK" goes.** On Home, **These go to channel** is normally **0 - Public**. Leave it unless your group agreed on another channel.

4. Using it: the Home screen

The **Home** screen is the only one you need in an emergency. Big buttons, plain words.

The colour bar at the top tells you if you can reach people

REACHABLE a repeater is heard. Your messages should get through.
NO REPEATER HEARD go higher or outside; radios right next to you can still hear you.
RADIO NOT CONNECTED plug the radio into the hub and press **Connect radio**.

The two big buttons

I'M OK Sends your tag (like `LC-K7M2P4XQ`) and your position. Press it when you arrive somewhere, and every hour or so. Or let the radio do it for you: **Power** → **Automatic I'M OK**.

I NEED HELP Asks first ("Send a call for help?"). If you say yes, the radio sends **NEEDS HELP** with your position, and **repeats it** every few minutes. The button turns into **CANCEL HELP CALL**. Cancelling tells everyone you are OK.

Call 911 first when you can.

CAIRN only reaches other radios. It does not call emergency services.

Send a status

One tap tells everyone what you need or are doing, with your position:

Need medical **Need water** **Need food** **Need shelter**
Need transport **Evacuating** **Staying put** **Can help**

Mark this place

Stand where something is, then tap what it is. It shows on every CAIRN map and in the report:

Water **Shelter** **Medical aid** **Hazard** **Meeting point**
Road blocked **Supplies**

Ready check

A list of ticks ✓ notes ! and problems ✕. Look at it when you set up and fix the red ones. It ends with **READY** when all is well.

Situation report

Press **Make a report** for a short summary of who has reported, who needs help and what is marked. You can read it aloud to a rescuer, copy it, save it, or send a one-line version to everyone.

When someone else calls for help

A big red box appears on every screen with a sound. It shows who, how long ago, and which direction and how far. Press **Show on map** to see where, and

Got it when someone is dealing with it. Turn the sound off with **Sound for help calls** on Home.

5. Reading what others send

The **People** screen lists everyone who has reported. It shows **NEEDS HELP** at the top, their last status, whether their battery is low, and where they are. Under **Needs and places** you see totals (for example “2 Need water”) and all marked places. **Overdue** means someone has not been heard from for a while (you can change how long).

The **Map** shows you, the repeaters, the other people, **red pulsing dots for help calls**, and orange diamonds for marked places. With no internet, load a map ahead of time (section 3).

6. Chat and channels

Open **Chat**. Pick a channel on the left, type your message, press **Send**. Messages are up to 140 letters. Remember that anyone in range may read the Public channel.

What is a channel?

A channel is a group. Everyone who has the same channel can read it, and everyone else cannot (except on Public and Open channels). There are three kinds:

Kind	What it is	Who can read it
Public	Built in. Channel 0.	Everyone with a CAIRN or MeshCore radio nearby
Open (starts with #)	Anyone who types the same name joins, for example `#water-tank`	Anyone who knows the name

Private	Needs a name **and** a secret key	Only people you give the key to

Add or remove a channel

In **Chat**, open **Channels: add or remove** (the grey bar under the people list).

1. **Join an open channel:** type the name (letters, numbers and dashes) in the first box and press **Add**.

2. **Make a private channel:** type a name and press **Make**. The radio makes the secret key and shows it. Give the **name and key** to your people (write it down or press **Copy**).

3. **Join a private channel:** type the name, then the 32-letter key you were given, and press **Join**.

4. **Remove a channel:** press **Remove** beside it, then **Yes, remove**. Public cannot be removed. The radio holds up to 40 channels.

Check-ins go to one channel

I'm OK, help calls, statuses and marks all go to the channel marked **check-ins go here**. It starts as Public so everyone nearby sees them. To use a private channel instead, press **Use for check-ins** beside it. Do this **only if everyone who might help you is on that channel**.

7. Standard and Disaster mode

	STANDARD	DISASTER

Use it	Every day, hikes, practice	A real emergency
Radio strength	The strongest the rules allow. Drops by itself when the battery is low (below 25%).	The strongest the rules allow, **always**, even on a low battery
Radio screen	Normal	Turns off sooner to save power
Quiet hours	Obeyed	Off, so nothing is silenced
Silent repeaters	Marked stale after a while	Marked stale sooner

Change it with the two buttons on **Home**, or on the radio's own **POWER** screen. The mode is remembered when the radio restarts. A running help call always uses full strength.

> **Why both modes have the same top power**
>
> In the United States, radios that need no licence have one legal limit (about 1 watt at the antenna, with an antenna up to 6 dBi) and there is no higher limit for emergencies. So Disaster mode does not make the signal stronger than Standard at full power. It makes sure the radio *stays* at full power and stops it saving battery.

8. Power: batteries, solar, and radio strength

What uses the power

The **tablet** uses the most (it is like a small lamp). The two radios together use little while listening. Sending a message costs very little. So: **turn the tablet screen down, close other apps, and let the tablet sleep**. The radios keep working while the tablet sleeps, and the CAIRN radio keeps the log.

The Power screen

- **Radio power and settings** — shows the strength now and why. **FULL** is the strongest allowed; **SAVER** uses less battery and reaches less far.
- **Battery saving** (on by default) — below 25% on battery, Standard mode uses the saver strength and automatic check-ins slow down. It turns itself off at 35% or when you plug power in.
- **Low-battery message** — at 15% the radio sends **LOW BATTERY** once so friends know you may go quiet.
- **How long will my power last?** — type in your power bank size, tablet use and solar panel. It tells you roughly how many hours or days you have. It is an estimate, not a promise.
- **Power banks timer** — a reminder. The radios cannot see how full a power bank is, so press the button when you start one.

Power banks and solar — simple rules

1. **Charge the power banks before you need them**, and top them up whenever you can.

2. **Put the solar panel in direct sun**, pointing at the sun, with nothing shading even a corner. Shade on part of a panel cuts its power a lot. Check it every hour or two and turn it toward the sun.

3. **Charge a power bank with the panel; run the kit from a power bank.** Charging and running at the same moment works only if your bank allows it.

4. **Keep batteries warm in the cold.** Cold makes them lose charge fast. Keep them inside a coat or bag.

5 **Use a hub and cables that can give enough power.** The radio draws short bursts of up to about 0.75 amp when it sends at full strength. A weak hub makes the radio restart. If it restarts when sending, change the hub or cable.

Antenna gain and the legal limit

The radio works out its power from the antenna you tell it about. The stock antenna is about 2 dBi. A bigger antenna (higher dBi) sends more strongly in one direction, so the rules make the radio send with *less* power. If you change the antenna, enter its number on the **Power** screen.

9. The radio's own little screen

The CAIRN radio shows a picture of a star over a stack of stones when it starts, then the home pages. **Short press** the button to go to the next page. **Hold** the button to do what the page says.

```
CAIRN STATION
LC-K7M2P4XQ
log 20KB 41 lines
check-in 4m ago
hold: send check-in
```

```
HELP
I NEED HELP call
repeats until cancelled

hold: arm
(then:) SEND HELP CALL?
Press again to send
```

```
STANDARD MODE
radio 28 dBm (631 mW)
battery 88% +power
full range
hold: switch mode
```

HELP page: hold once to arm ("SEND HELP CALL?"), then hold again within 5 seconds to send. This stops accidental calls in a pocket. While the call is

running, hold once to cancel. **POWER page:** hold to switch between Standard and Disaster.

10. If something is not working

What you see	What to do
Cannot press **Connect radio**, or the radio is not in the list	Use **Chrome** (version 148 or newer on Android). Unplug the radio and plug it in again. Close other apps that use USB. Use a cable that carries data, not a charge-only cable.
NO REPEATER HEARD	Go outside, up high, away from metal and walls. Check that the antenna is tight. Wait 10 minutes. Radios near you can still hear you.
Buttons are grey	The radio is not connected yet. Press **Connect radio**.
“No position yet” (no GPS)	Go outside or stand by a window and wait a few minutes. Messages still work; they just say *no GPS fix*.
The radio keeps restarting when sending	The hub or cable cannot give enough power. Plug the radio into a stronger port or power bank, or use a shorter cable.
Power seems low	See section 8. Check **Battery saving**, and that the antenna number is right.
Chat says the person’s key has not been heard	Wait for their radio to announce itself (it can take several minutes), then try again.
Typing is hard	Use the little keyboard button beside a box to bring up the on-screen keyboard, or the **Keyboard** button at the top.

Everything looks wrong	Use the radio's small **reset**: unplug it, wait 10 seconds, plug it in. Your channels, mode and settings are remembered.

11. Safety, privacy and the rules

Emergency services

If you can use a phone, **call 911** (or your local number). CAIRN is a backup. It cannot reach 911.

Privacy

Your tag and **position** go out to everyone in range when you press I'M OK, a status, a mark, or a help call. Messages on Public and Open channels can be read by anyone. Private channels are for people who have the key.

- **Antennas on first.** Never send with no antenna.
- **Do not stand close to the antenna while sending.** Keep the antenna a little away from your body and from other people when you are sending a lot.
- **No licence needed in the US.** CAIRN is designed to stay inside the limit for unlicensed (FCC Part 15) radios in the 902–928 MHz band. For that to be true, use only the antenna the radio expects (or enter the right number). You may not cause harmful interference, and you must accept interference from others.
- **Outside the US,** follow your own country's rules for this kind of radio, and check the radio's frequency setting.
- **This is ALPHA software.** It is still being tested. Practise with it before you rely on it. It is a help, not a guarantee.

12. Commands (for helpers)

You never need these to use CAIRN. They can be typed into the **Console** (Full view → Station) or a serial terminal. Every button above sends one of them.

Type	What it does
`cairn`	Station tag, mode, journal size
`cairn ok [channel]`	Send I'M OK (to the check-in channel unless a number is given)
`cairn help [channel]` / `cairn help off`	Start or cancel the repeating help call
`cairn beacon 5`	Help call repeats every 5 minutes (1–60)
`cairn ustatus 2`	Send status number 2 (1 Need medical, 2 Need water, 3 Need food, 4 Need shelter, 5 Need transport, 6 Evacuating, 7 Staying put, 8 Can help)
`cairn mark 4`	Mark this place: 1 Water, 2 Shelter, 3 Medical aid, 4 Hazard, 5 Meeting point, 6 Road blocked, 7 Supplies
`cairn mode standard` / `cairn mode disaster`	Change mode
`cairn power full` / `saver` / `status`	Power choice, or show the power now and the limit
`cairn antenna 2`	Tell the radio the antenna gain in dBi (0–20)
`cairn auto 30` / `cairn auto off`	Automatic I'M OK every 30 minutes (10–1440), or off

`cairn eco on` / `off`, `cairn lowbat on` / `off`	Battery saving; low-battery message
`cairn main 0`	Channel used for check-ins, help, status and marks
`cairn chans`	List channels
`cairn chan add #name`	Join an open channel
`cairn chan new Name`	Make a private channel with a new key
`cairn chan add Name <32-character key>`	Join a private channel
`cairn chan key 3` / `cairn chan remove 3`	Show channel 3's key / remove channel 3
`cairn say 0 hello`	Send a message to channel 0
`cairn config`	Show all settings
`cairn dump` / `cairn clear yes` / `cairn time <seconds>`	Send the log / erase the radio's log / set the radio clock

13. Kit checklist

Tick these off before you go. Print this page and keep it in the kit.

☐ Tablet, charged	☐ USB hub	☐ CAIRN radio + antenna

☐ Repeater radio + antenna	☐ Power banks (charged)	☐ Solar panel + cable
☐ Spare cables (data cables)	☐ Offline map loaded	☐ This guide on paper
☐ Antenna gain number entered	☐ Ready check says READY	☐ Everyone knows the check-in channel

14. Words used in this guide

Station tag	Your radio's name tag, like `LC-K7M2P4XQ`. It is made once and never changes.
Repeater	A radio whose job is to pass messages along so they go farther.
Channel	A group chat. Everyone with the same channel reads it.
Key	A secret code (32 letters and numbers) that opens a private channel.
GPS fix	The radio knows where it is, from satellites. It needs a view of the sky.
dBi	How strongly an antenna focuses the signal. The stock antenna is about 2.
mW / W	Milliwatts and watts: how strongly the radio sends. 1000 mW is 1 W.
Overdue	Someone has not been heard from for longer than you set.

Engineered by: Aries Thorne & Todd Steinhauer for Wolf Reactive Systems (WRS)
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Scates for their guidance and support.

LODESTAR CAIRN is ALPHA software. Mesh radio is not a replacement for 911, a phone or a satellite messenger.